Chapter 2: Boolean Algebra and Logic Gates

- Boolean algebra, like any other deductive mathematical system, may be defined with a set of elements, a set of operators, and a number of unproved axioms or postulates.
- A $\boxed{\text{binary operator}}$ defined on a set S of elements is a rule that assigns to each pair of elements from S a unique element from S.
- As an example, consider the relation $a*b=c$. We say that $*$ is a binary operator if it specifies a rule for finding c from the pair (a,b) and also if $a,b,c \in S$. However, $*$ is not a binary operator if $a,b \in S$, while the rule finds $c \notin S$.
- The postulates of a mathematical system form the basic assumptions from which it is possible to deduce the rules, theorems, and properties of the system.

- The most common postulates used to formulate various algebraic structures are:

1. $\boxed{\text{Closure:}}$ A set S is closed with respect to a binary operator if, for every pair of elements of S, the binary operator specifies a rule for obtaining a unique element of S.

- For example, the set of natural numbers $N=\{1,2,3,4,\dots\}$ is closed with respect to the binary operator plus $(+)$ by the rules of arithmetic addition, since for any $a,b \in N$ we obtain a unique $c \in N$ by the operation $a+b=c$.
- The set of natural numbers is not closed with respect to the binary operator $(-)$ by the rules of arithmetic subtraction because $2-3=-1$ and $2,3 \in N$, while $(-1) \notin N$.

2. $\boxed{\text{Associative law:}}$ A binary operator $*$ on a set S is said to be associative whenever: $(x*y)*z = x*(y*z)$ for all $x,y,z \in S$.

3. $\boxed{\text{Commutative law:}}$ A binary operator $*$ on a set S is said to be commutative whenever: $x*y = y*x$ for all $x,y \in S$.

4. $\boxed{\text{Identity element:}}$ A set S is said to have an identity element with respect to a binary operation $*$ on S if there exists an element $e \in S$ with the property:

$$e*x = x*e = x \quad \text{for every } x \in S.$$

$\boxed{\text{Example:}}$ The element 0 is an identity element with respect to operation $+$ on the set of integers $I=\{\dots,-3,-2,-1,0,1,2,3,\dots\}$ since:

$$x+0 = 0+x = x \text{ for any } x \in I.$$